

PROJECT REPORT

ON

“ComicCraft – AI Comic Story Creator using Gemini Models”

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of
BACHELOR OF COMPUTER APPLICATIONS (BCA)

Submitted By

- | | |
|--------------------------------|--------------------|
| 1. Team Leader – SAKTHIVELAN P | Reg. No: 212506758 |
| 2. Team Member – RITHISH D | Reg. No: 212506757 |

Under the Guidance of
P SAKTHI VELAN

Department of Computer Applications

TJS COLLEGE OF ARTS AND SCIENCE
Academic Year 2026–2027

CERTIFICATE

This is to certify that the project entitled “ComicCraft – AI Comic Story Creator using Gemini Models” is a bonafide work carried out by the above-mentioned students in partial fulfillment of the requirements for the award of the Bachelor of Computer Applications (BCA) degree.

The project has been completed under my guidance and supervision during the academic year 2026–2027.

Project Guide

P SAKTHI VELAN

Head of the Department

V SIVARANJANI

DECLARATION

We hereby declare that the project report entitled “ComicCraft – AI Comic Story Creator using Gemini Models” is our original work carried out as part of the Bachelor of Computer Applications programme.

We further declare that this project has not been submitted previously, either fully or partially, to any other institution or university for the award of any degree or qualification.

Team Leader

Name: SAKTHIVELAN P

Register No: 212506758

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported us in completing the project entitled “ComicCraft – AI Comic Story Creator using Gemini Models.”

We thank the management and the Department of Computer Applications for providing the opportunity and facilities required to complete this project.

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We also thank all teaching and non-teaching staff members, friends and family members for their continuous support and encouragement.

Finally, we thank everyone who contributed directly or indirectly to the successful completion of this project.

ABSTRACT

ComicCraft is an AI-powered application designed to help users create comic stories with the assistance of Gemini models. The system allows users to provide a story idea, select suitable preferences and generate a structured comic storyline.

The application can use generative AI to create characters, scenes, dialogues and story sequences from a user's prompt. The generated content can then be organized into comic panels and displayed in a simple, user-friendly interface.

ComicCraft aims to reduce the time and effort required to develop an original comic concept. It can be useful for students, writers, hobby creators and users interested in experimenting with AI-assisted storytelling.

The system focuses on creative assistance. Generated stories and images should be reviewed by users before publication, especially when originality, copyright or other content requirements are involved.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Comic creation traditionally involves planning a storyline, developing characters, writing dialogues and arranging scenes into panels. These activities require creativity as well as considerable time and effort.

1.2 Problem Statement

Many beginners have difficulty converting a simple story idea into a complete comic structure. Creating characters, scenes and dialogues manually can be time-consuming.

1.3 Objectives

- To develop an AI-assisted platform for creating comic stories.
- To convert user ideas into structured comic storylines.
- To generate characters, scenes and dialogues using Gemini models.
- To organize generated content into comic panels.
- To provide an easy interface for editing and previewing the story.

1.4 Scope

The system can be extended to support different genres, character styles, languages, panel layouts, image generation and export formats.

CHAPTER 2 – EXISTING SYSTEM

2.1 Existing Method

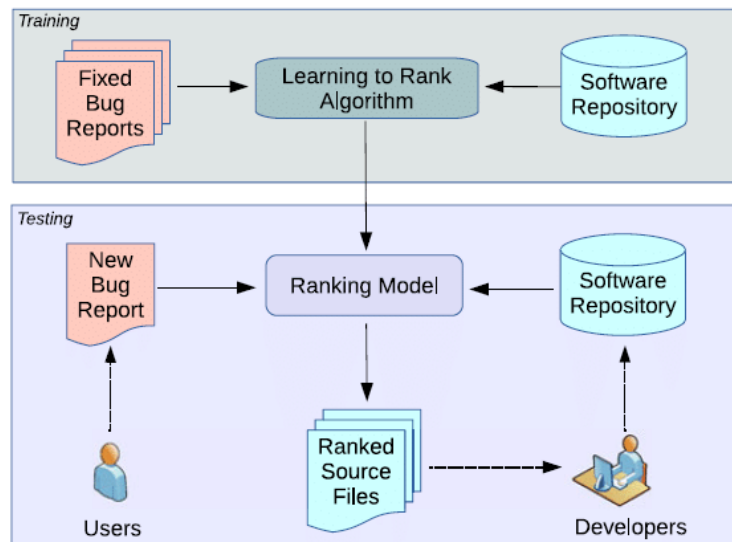
In the traditional approach, comic creators plan stories manually, design characters, write dialogues and arrange panels using separate tools. Beginners may need several applications to complete a single comic.

2.2 Limitations

- Manual story development requires more time.
- Creating consistent characters can be difficult.
- Panel planning and dialogue writing require repeated effort.
- Separate tools may be needed for writing and visual design.
- Beginners may find it difficult to structure a complete comic.

2.3 Need for AI Assistance

Generative AI can assist users by producing story ideas, character descriptions, scene descriptions and dialogues from natural-language prompts. This can make the creative workflow faster and more accessible.



CHAPTER 3 – PROPOSED SYSTEM

3.1 Proposed System

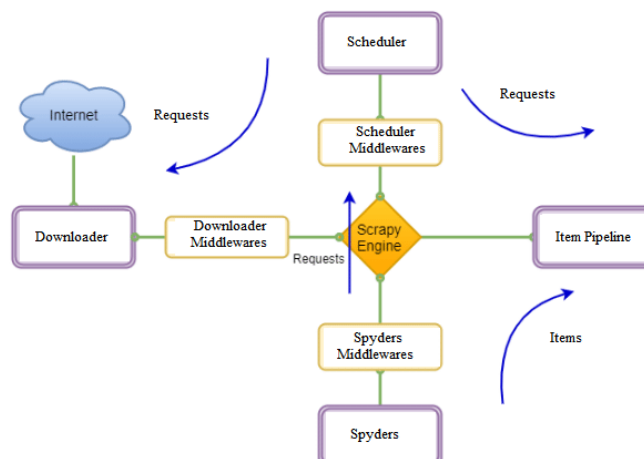
ComicCraft combines a web interface with Gemini-based generative AI to assist in comic story creation. The user provides a concept and the system converts it into a structured story with characters, scenes, dialogues and panels.

3.2 Main Features

- User prompt and story idea input.
- Genre and style selection.
- AI-generated story outline.
- Character generation and descriptions.
- Scene and dialogue generation.
- Comic panel organization.
- Preview and editing.
- Save and export functionality.

3.3 Proposed Workflow

User Input → Prompt Processing → Gemini Model → Story and Character Generation → Panel Structuring → Preview → Edit / Save / Export



CHAPTER 4 – SYSTEM REQUIREMENTS

4.1 Hardware Requirements

- Processor: Dual Core processor or above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Display: 1366 × 768 or higher
- Internet connection for Gemini API features

4.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Frontend: HTML, CSS, JavaScript or React
- Backend: Python Flask / Django or Node.js
- Database: MySQL / PostgreSQL / SQLite
- AI Platform: Gemini API / Google AI services
- Browser: Chrome, Edge or Firefox

4.3 User Requirements

Users need basic knowledge of web applications and should provide clear prompts describing the comic they want to create.



CHAPTER 5 – SYSTEM ANALYSIS

5.1 Functional Requirements

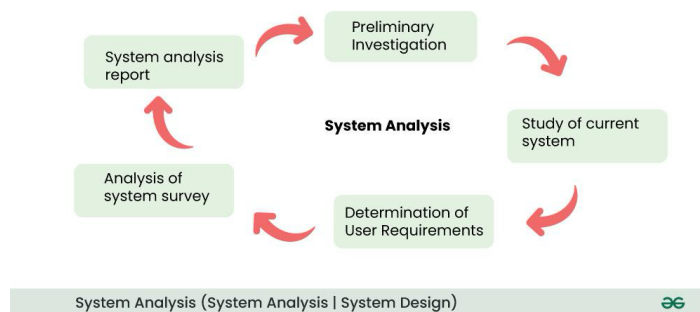
- The system shall accept a comic story prompt.
- The system shall allow genre and style selection.
- The system shall generate a story outline.
- The system shall generate character and scene descriptions.
- The system shall generate dialogues for panels.
- The system shall display the generated comic structure.
- The system shall allow saving and exporting.

5.2 Non-Functional Requirements

- Usability – simple and understandable interface.
- Performance – reasonable response time.
- Security – protect user information and API credentials.
- Reliability – handle invalid prompts and API errors.
- Maintainability – allow new features and templates.

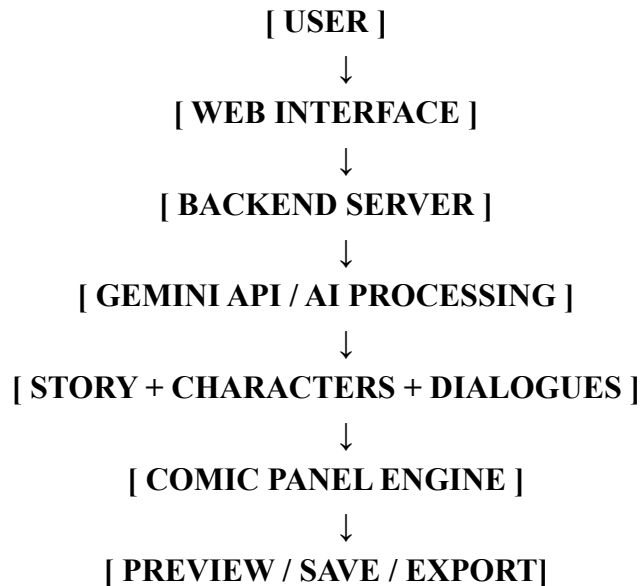
5.3 Feasibility

The project is technically feasible because web technologies and cloud-based generative AI APIs can be integrated to create an AI-assisted storytelling application.



CHAPTER 6 – SYSTEM DESIGN

6.1 System Architecture

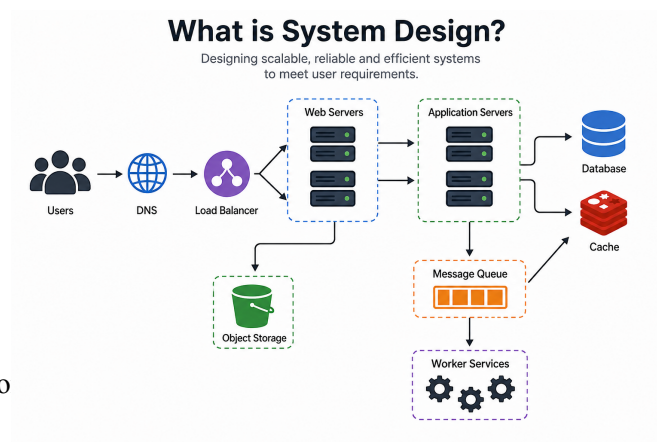


6.2 Use Case Description

- User enters a comic idea.
- User selects genre and style.
- System sends the prompt for AI processing.
- Gemini model generates story content.
- System organizes content into panels.
- User previews and edits the comic.
- User saves or exports the final result.

6.3 Data Flow

Input from the user is validated and converted into a structured prompt. The backend communicates with the Gemini model and receives generated content. The application then formats the response into characters, scenes, dialogues and comic panels.



CHAPTER 7 – MODULES

7.1 User Module

Manages user registration, login and basic profile information.

7.2 Story Input Module

Collects the user's story idea, genre, theme, language and preferred style.

7.3 AI Generation Module

Sends suitable prompts to Gemini models and processes the generated story content.

7.4 Character Module

Creates character names, roles, personalities and visual descriptions.

7.5 Panel Module

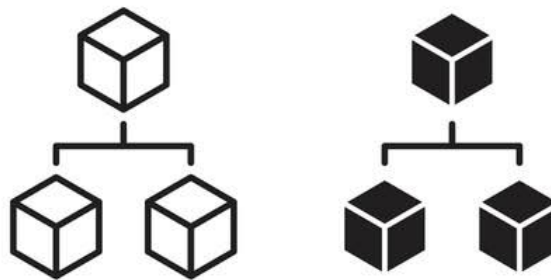
Breaks the story into scenes and organizes each scene as a comic panel with dialogue and descriptions.

7.6 Preview and Editing Module

Allows users to review generated content and make changes before saving.

7.7 Export Module

Provides options for saving or exporting the completed comic project.



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CHAPTER 8 – TECHNOLOGIES USED

8.1 Frontend

HTML, CSS and JavaScript can be used to create the interface. React can be used for a component-based and interactive application.

8.2 Backend

Python Flask/Django or Node.js can manage requests, authentication, prompt processing and communication with AI services.

8.3 Gemini Models

Gemini models can be used for generative AI tasks such as story creation, character descriptions, scene descriptions and dialogue generation. The exact model used can depend on the available API and project requirements.

8.4 Database

SQLite, MySQL or PostgreSQL can store users, projects, prompts, generated stories and saved comic information.

8.5 Document / Image Processing

Additional libraries or APIs can be integrated for rendering comic panels, creating images and exporting the final comic.



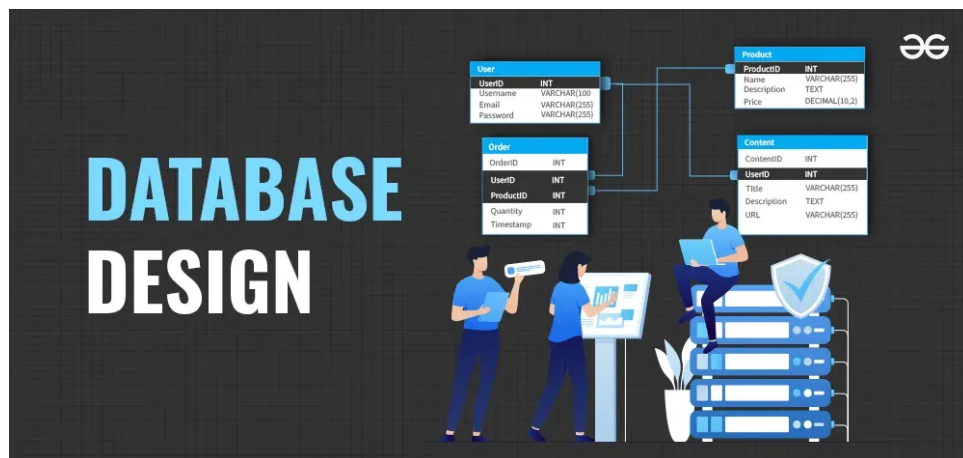
CHAPTER 9 – DATABASE DESIGN

Table	Field	Type	Purpose
Users	user_id	Integer	Unique user ID
Users	name	Varchar	User name
Users	email	Varchar	Login email
Projects	project_id	Integer	Unique comic project ID
Projects	title	Varchar	Comic title
Projects	genre	Varchar	Comic genre
Projects	prompt	Text	Original user prompt
Projects	content	Text	Generated story content
Projects	created_at	DateTime	Project creation date

9.1 Database Relationship

One user can create multiple comic projects. Therefore, the Users table can have a one-to-many relationship with the Projects table through user_id.

Users (1) ————— (M) Projects



CHAPTER 10 – IMPLEMENTATION

10.1 Implementation Steps

- Create the frontend and backend environment.
- Create database tables for users and projects.
- Develop story input and project creation screens.
- Configure secure Gemini API access.
- Create prompts for story, character and panel generation.
- Process and structure Gemini responses.
- Display the comic panels in the interface.
- Implement editing, saving and export features.
- Test the complete application.

10.2 Sample Workflow

A user enters: “Create a short adventure comic about three friends discovering a mysterious forest.” The application can generate a title, characters, story outline, scenes and dialogues. These outputs are then organized into panels for preview.

10.3 Security Considerations

API keys should be stored securely on the server and should not be exposed in frontend code. User data should be protected through authentication, validation and controlled database access.



CHAPTER 11 – TESTING

11.1 Testing Objective

Testing ensures that each module works correctly and that the application handles invalid input, API failures and incomplete data.

Test Case	Input / Action	Expected Result	Status
TC01	Valid user login	Dashboard displayed	Pass
TC02	Empty story prompt	Validation message displayed	Pass
TC03	Valid story prompt	AI-generated story returned	Pass
TC04	Select genre	Selected genre applied	Pass
TC05	Generate characters	Character details displayed	Pass
TC06	Generate panels	Story divided into panels	Pass
TC07	Preview project	Comic preview displayed	Pass
TC08	Save project	Project stored successfully	Pass

11.2 Testing Types

The project can use unit testing, integration testing, system testing and user acceptance testing.



CHAPTER 12 – RESULTS AND SCREENSHOTS

The proposed ComicCraft application provides an AI-assisted workflow for transforming a simple story idea into a structured comic. The user can generate story content, characters, scenes and dialogues and then view them as comic panels.

12.1 Expected Screens

- Login / Registration Screen
- ComicCraft Dashboard
- Story Prompt Screen
- Genre and Style Selection
- AI Generation Screen
- Character and Scene Screen
- Comic Panel Preview
- Save / Export Screen

[SCREENSHOT PLACEHOLDER – DASHBOARD]

[SCREENSHOT PLACEHOLDER – STORY GENERATION]

[SCREENSHOT PLACEHOLDER – COMIC PANEL PREVIEW]

Actual screenshots can be inserted here after implementing the application.

CHAPTER 13 – ADVANTAGES, LIMITATIONS & FUTURE SCOPE

13.1 Advantages

- Reduces time required for initial comic development.
- Helps beginners convert ideas into structured stories.
- Generates characters and dialogues quickly.
- Provides a single workflow for story and panel planning.
- Can support different genres and styles.

13.2 Limitations

- AI output may require editing and verification.
- Generated content may not always match the user's exact intention.
- API availability and usage limits may affect generation.
- Generated images or content may require copyright and originality review before publication.

13.3 Future Enhancements

- Multilingual comic generation.
- More advanced character consistency.
- AI-generated comic artwork.
- Voice narration and sound effects.
- Collaborative comic editing.
- PDF and image export.
- Character library and reusable templates.



CHAPTER 14 – CONCLUSION AND REFERENCES

14.1 Conclusion

ComicCraft demonstrates how generative AI can assist the creative process of comic storytelling. By combining a user-friendly interface with Gemini models, the system can transform a basic idea into a structured story containing characters, scenes, dialogues and comic panels.

The project can be further enhanced with image generation, character consistency, multilingual support, collaboration and advanced export options. ComicCraft is intended as a creative assistance tool, with users responsible for reviewing generated content before publication or other use.

14.2 References

- Google AI for Developers – Gemini API documentation
- Google Gemini documentation and developer resources
- MDN Web Docs – <https://developer.mozilla.org/>
- React Documentation – <https://react.dev/>
- Python Documentation – <https://docs.python.org/>
- MySQL Documentation – <https://dev.mysql.com/doc/>

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